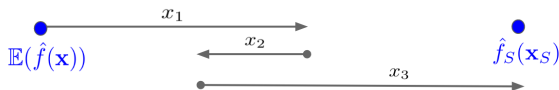


FROM GAME THEORY TO MACHINE LEARNING

- Game: Make prediction $\hat{f}(x_1, x_2, \dots, x_p)$ for a single observation $\mathbf{x}$
- Players: Features $x_j, j \in \{1, \dots, p\}$ which cooperate to produce a prediction
 - $\rightsquigarrow$ How can we make a prediction with a subset of features without changing the model?
 - $\rightsquigarrow$ PD function: $\hat{f}_S(\mathbf{x}_S) := \int_{\mathcal{X}_{-S}} \hat{f}(\mathbf{x}_S, \mathbf{x}_{-S}) d\mathbb{P}_{\mathbf{x}_{-S}}$ ("removing" by marginalizing over $-S$)
- Value function / payout of coalition $S \subseteq P$ for observation $\mathbf{x}$:

$$v(S) = \hat{f}_S(\mathbf{x}_S) - \mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x})), \text{ where } \hat{f}_S : \mathcal{X}_S \mapsto \mathcal{Y}$$

$\rightsquigarrow$ subtraction of $\mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ ensures that v is a value function with $v(\emptyset) = 0$



- Marginal contribution: $v(S \cup \{j\}) - v(S) = \hat{f}_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - \hat{f}_S(\mathbf{x}_S)$
 - $\rightsquigarrow \mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ cancels out due to the subtraction of value functions